

Single-cell transcriptomics reveals a cross-cohort decidual inflammatory macrophage/APC signature in recurrent pregnancy loss

Short title: Cross-cohort decidual inflammatory macrophage/APC signature in RPL

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Reuben S. Maghembe^{1,2*}

¹Department of Microbiology and Parasitology, Faculty of Medicine, St. Francis University College of Health and Allied Sciences, Ifakara, Tanzania

²Department of Omics and Computational Biology, AfroBiomix Co. Ltd., Bridge Street, Samora Tower, Kivukoni, Dar es Salaam, Tanzania

*Corresponding author: Reuben S. Maghembe

Email: rmaghembe@gmail.com; rmaghembe@sfuchas.ac.tz

Abstract

Recurrent pregnancy loss is a clinically heterogeneous reproductive disorder and a major reproductive-health problem. A substantial fraction of affected patients remain without a definitive explanation after evaluation for chromosomal, anatomical, endocrine, thrombotic and immune contributors, while public transcriptomic studies remain difficult to compare because decidual bulk profiles mix cell abundance, cell state and sampling differences. Here, we used a single-cell-informed transcriptomic transportability design to test whether recurrent-miscarriage-associated tissue-state programmes derived from a disease single-cell/pseudobulk reference recur across independent bulk decidua cohorts. GSE214607 was used to derive compact projection-ready programmes from first-trimester decidua and villi samples. Four signatures were projected into three independent bulk decidua datasets, GSE161969, GSE113790 and GSE178535, using within-cohort standardized expression and mean programme scoring. The primary recurrent-miscarriage decidual inflammatory macrophage/APC programme showed complete 30-gene coverage in all three external cohorts and positive case-associated shifts in each dataset. Cohort-level deltas were +1.019, +0.282 and +0.941, giving a mean delta of +0.747 and positive directional recurrence in 3/3 cohorts. Secondary decidua immune-stromal, villi trophoblast/epithelial and recurrent-miscarriage villi structural/trophoblast programmes showed weaker, modest or context-dependent behaviour. These findings support exploratory programme-level transportability of an inflammatory decidual macrophage/APC tissue-state axis in recurrent pregnancy loss. The result is observational and hypothesis-generating; it does not establish causality, diagnostic performance, clinical utility or standalone biomarker validation.

Introduction

Infertility and pregnancy loss impose substantial clinical, psychological and societal burdens. The World Health Organization estimates infertility as a common global reproductive-health condition, while recurrent pregnancy loss (RPL) represents a distinct clinical problem requiring its own evaluation and management pathway [1-3]. Contemporary guidelines differ in definitions and thresholds, but both ESHRE and ASRM emphasize the heterogeneity of RPL and the need to evaluate established contributors such as genetic abnormalities, uterine factors, endocrine disease, antiphospholipid syndrome and other maternal or embryonic causes [2,3]. Beyond these recognized categories, miscarriage and recurrent miscarriage also carry physical, psychological and future obstetric consequences, making mechanistic understanding of early pregnancy failure a human-health priority rather than a narrow fertility issue [4,5].

41 The early maternal-fetal interface provides a biologically plausible site for unexplained or incompletely
42 explained pregnancy loss. Human decidualization transforms endometrial stromal fibroblasts into specialized
43 decidual cells, coordinates glandular and vascular remodelling, and creates a tissue matrix in which immune
44 competence must coexist with tolerance of fetal trophoblast [6,7]. Single-cell atlases of first-trimester decidua
45 and placenta have shown that this interface is a structured multicellular ecosystem containing decidual stromal
46 cells, macrophages, natural killer cells, T cells, endothelial and mural cells, epithelial compartments and
47 invading trophoblasts [8,9]. Trophoblast-decidual immune crosstalk, local trophoblast recognition by uterine
48 immune cells and regulatory T-cell programmes are therefore central to early placentation and immune balance
49 [10-13].

50 RPL has repeatedly been linked to immune dysregulation, but the direction, cellular basis and translational
51 meaning of immune signals remain unresolved. Single-cell and immune-profiling studies have reported altered
52 distributions of decidual leukocyte subsets in RPL, unexplained RPL and recurrent miscarriage, including
53 changes in NK-cell, T-cell and myeloid compartments [14-17]. Decidual macrophages are particularly relevant
54 because they participate in phagocytosis, tissue remodelling, angiogenesis, trophoblast interaction, cytokine
55 production and immune regulation [18,22,23]. Disease-focused studies have associated RPL or recurrent
56 spontaneous abortion with altered decidual macrophage states, CCR1+ macrophage dysfunction, M1-like
57 polarization and macrophage-related diagnostic signatures [18-21,30]. Complement-associated macrophage
58 biology is also plausible at the maternal-fetal interface because C1q and complement-related pathways
59 contribute to implantation-site inflammation, trophoblast invasion and tissue remodelling, although excessive or
60 mislocalized complement activity can be harmful [24-27].

61 Despite this biological rationale, public transcriptomic evidence remains hard to integrate. Bulk decidual RNA
62 profiles can detect disease-associated expression differences, but they cannot directly distinguish increased
63 macrophage abundance from macrophage activation, altered decidual stromal composition, trophoblast
64 admixture, tissue injury or platform-specific artefact. Many public RPL studies are small, and single-cohort
65 differential-expression lists may not transport across datasets [31-36]. Single-cell-informed projection offers an
66 intermediate strategy: derive interpretable programmes from a cellular reference, then test whether those
67 programmes recur directionally across independent bulk cohorts. This approach is different from diagnostic
68 classifier construction; it evaluates whether a biologically coherent tissue-state axis is detectable beyond the
69 discovery dataset.

70 The present study addresses this gap by testing the transportability of GSE214607-derived recurrent-miscarriage
71 and tissue-context programmes across three independent bulk decidua cohorts. The aims were: first, to derive
72 compact and interpretable projection-ready programmes from a disease single-cell/pseudobulk reference;
73 second, to determine whether these programmes show adequate gene coverage when projected into independent
74 bulk decidua datasets; and third, to assess whether a recurrent-miscarriage decidual inflammatory
75 macrophage/APC programme shows concordant case-associated recurrence across cohorts. Figure 1 summarizes
76 the workflow, Table 1 summarizes the datasets, and Supplementary Table S1 documents metadata inclusion.

77 **Materials and Methods**

78 This was a retrospective computational reanalysis of public de-identified transcriptomic datasets. No new
79 participants were recruited, and no new identifiable human data were generated. The analysis was designed
80 around programme transportability rather than biomarker discovery or diagnostic classification. The overall
81 workflow is summarized in Fig. 1.

82 GSE214607 was used as the primary disease single-cell reference because it contains first-trimester decidua and
83 villi samples from control and recurrent-miscarriage groups and was originally generated to explore

84 macrophage-associated recurrent miscarriage signals [28-30]. Raw 10x Genomics matrix files were imported as
85 sparse count matrices, checked for matrix/barcode/feature consistency and assigned to control decidua,
86 recurrent-miscarriage decidua, control villi or recurrent-miscarriage villi categories. Per-cell quality-control
87 metrics included total counts and detected genes. Cells passing predefined feature thresholds were retained for
88 sample-level aggregation.

89 To avoid pseudoreplication, retained GSE214607 cells were aggregated into sample-level pseudobulk profiles
90 by summing counts within each sample. Library-size-normalized log2 counts per million were then calculated.
91 Pseudobulk aggregation was used for programme derivation because cells from the same individual are not
92 independent biological replicates and single-cell differential-expression procedures can underestimate variance
93 if sample structure is ignored [37,38]. The pseudobulk profiles were used only to derive exploratory
94 programmes and not to make formal differential-expression claims.

95 Candidate programmes were derived from three contrast families: recurrent-miscarriage decidua versus control
96 decidua, recurrent-miscarriage villi versus control villi and decidua versus villi. Candidate genes were ranked by
97 direction and magnitude of expression shift. Lists were then filtered for interpretability: low-information
98 noncoding, housekeeping, ribosomal, mitochondrial and technical features were excluded unless they had
99 explicit maternal-fetal-interface relevance. The final projection-ready framework retained four 30-gene
100 signatures: a recurrent-miscarriage decidual inflammatory macrophage/APC programme, a decidua immune-
101 stromal identity programme, a villi trophoblast/epithelial identity programme and a recurrent-miscarriage villi
102 structural/trophoblast programme (Table 2; Supplementary Table S2).

103 Three external bulk decidua datasets were used for projection: GSE161969, GSE113790 and GSE178535 [31-
104 36]. GSE161969 was processed from a public gene-symbol FPKM matrix and transformed as $\log_2(\text{FPKM} + 1)$.
105 GSE113790 was imported as a processed FPKM expression matrix and curated at the gene-symbol level.
106 GSE178535 was analysed from its public normalized log2 expression matrix. Sample labels were manually
107 curated to harmonize control and RPL/RSA case categories while preserving dataset-specific terminology
108 (Supplementary Table S1).

109 For each cohort and signature, genes were intersected with the cohort expression matrix, and coverage was
110 recorded before scoring. Within each cohort, each gene was standardized across samples. Programme scores
111 were then calculated as the mean standardized expression of the matched signature genes. This simple mean-z
112 scoring was chosen to preserve interpretability and avoid cross-platform absolute-expression assumptions;
113 related single-sample gene-set approaches use analogous logic to summarize coordinated expression at the
114 sample level [39].

115 Transportability was evaluated using technical coverage, case-minus-control programme-score deltas and
116 directional concordance. Complete or near-complete signature coverage was required for confident
117 interpretation. For each signature and dataset, mean scores were calculated for controls and cases, and case-
118 minus-control deltas were computed. Directional recurrence was defined as a consistent sign of case-associated
119 shift across cohorts. Wilcoxon rank-sum tests and Benjamini-Hochberg-adjusted values were calculated
120 descriptively but were not used as the primary criterion because the external cohorts were small [40]. Full
121 projection-score and metadata files are listed in Supplementary Tables S2-S4.

122 Results

123 The analysis integrated one disease single-cell/pseudobulk reference and three independent bulk decidua
124 cohorts. GSE214607 provided the programme-derivation reference, while GSE161969, GSE113790 and
125 GSE178535 provided independent projection settings (Table 1; Fig. 1). The use of three external cohorts

126 allowed the study to prioritize directional recurrence across independent datasets rather than within-dataset
127 statistical significance.

128 Pseudobulk contrasts from GSE214607 generated both recurrent-miscarriage-associated and tissue-context
129 programmes. After interpretability filtering, four compact signatures were retained (Table 2). The recurrent-
130 miscarriage decidual inflammatory macrophage/APC programme was designated the primary programme
131 because it was derived from the disease-associated decidual contrast and included coherent myeloid, antigen-
132 presentation and complement-associated genes. The signature included HLA-DRA, HLA-DPA1, HLA-DPB1,
133 CD74, S100A8, S100A9, AIF1, MS4A7, C1QA, C1QB, C1QC, CTSZ, SPP1 and GRN, supporting
134 interpretation as a macrophage/APC-rich inflammatory tissue-state programme rather than an arbitrary gene list
135 (Table 2; Supplementary Table S2; Fig. 2).

136 The primary macrophage/APC programme showed complete technical coverage in all external bulk cohorts. All
137 30 genes were detected in GSE161969, GSE113790 and GSE178535, supporting reliable projection across
138 different processed expression matrices. Secondary signatures also had high coverage, but the macrophage/APC
139 programme was the most technically robust primary disease-associated signature (Supplementary Table S3;
140 Supplementary Fig. 1).

141 The macrophage/APC programme showed a positive case-associated shift in all three external cohorts. The
142 case-minus-control delta was +1.019 in GSE161969, +0.282 in GSE113790 and +0.941 in GSE178535. This
143 yielded positive directional recurrence in 3/3 cohorts, a mean delta of +0.747 and a median delta of +0.941
144 (Table 3; Fig. 3). Because the cohorts were small, this result is interpreted as exploratory directional recurrence
145 rather than a validated diagnostic effect. Nevertheless, the consistency of direction across three datasets supports
146 programme-level transportability of the decidual inflammatory macrophage/APC axis.

147 Secondary signatures showed weaker or context-dependent behaviour. The decidua immune-stromal identity
148 programme was positive in all three cohorts but with smaller deltas (+0.573, +0.206 and +0.031). The villi
149 trophoblast/epithelial identity programme showed modest positive shifts (+0.134, +0.406 and +0.156), while the
150 recurrent-miscarriage villi structural/trophoblast programme was mixed (-0.097, +0.245 and -0.007). Thus, not
151 all GSE214607-derived signatures showed the same behaviour in decidua-focused external cohorts (Table 3;
152 Fig. 4). This pattern supports tissue-context-aware interpretation and argues against a nonspecific projection
153 artefact shared equally by all signatures.

154 Together, the results identify a single-cell-derived recurrent-miscarriage decidual inflammatory
155 macrophage/APC programme with complete 30-gene coverage and positive directional recurrence across three
156 independent bulk decidua cohorts. The main result is a reproducible tissue-state signal, not a causal proof,
157 clinical diagnostic model or biomarker validation. Intermediate single-cohort and two-cohort analyses are
158 retained for reproducibility in Supplementary Fig. 2.

159 Discussion

160 This study provides exploratory evidence that a GSE214607-derived recurrent-miscarriage decidual
161 inflammatory macrophage/APC programme is directionally transportable across three independent bulk decidua
162 cohorts. The study was designed to answer a conservative reproducibility question: does an interpretable single-
163 cell-derived tissue-state programme recur across datasets, tissues and preprocessing contexts? The answer was
164 strongest for the macrophage/APC programme, which showed complete gene coverage and positive case-
165 associated shifts in GSE161969, GSE113790 and GSE178535. This directly addresses the first and second aims
166 by demonstrating that compact programme derivation and technically robust projection are feasible, and
167 addresses the third aim by showing concordant cross-cohort recurrence of the primary inflammatory myeloid
168 programme.

Scientifically, the result fits a growing literature in which RPL is viewed not only as an embryonic or anatomical problem but also as a disorder that may involve disturbed decidual immune balance. Guidelines emphasize that RPL is clinically heterogeneous and that many cases remain unexplained after standard evaluation [2,3]. The Lancet miscarriage series further frames recurrent miscarriage as a high-burden condition that requires better biological understanding and improved care pathways [4,5]. The present data do not redefine RPL clinically, but they add a reproducible transcriptomic tissue-state observation that may help organize future mechanistic work.

The macrophage/APC programme is biologically plausible because macrophages are positioned at the interface between immune tolerance, tissue repair, trophoblast interaction, angiogenesis and inflammation. Single-cell studies of normal early pregnancy show that decidua contains diverse immune and stromal populations whose interactions are essential for successful placentation [8-11]. Disease-focused single-cell and immune-profiling studies have independently reported altered decidual leukocyte states in RPL or unexplained RPL, including myeloid, NK-cell and T-cell alterations [14-17]. The current finding is therefore similar in direction to published immune-landscape studies: it supports the presence of an altered decidual immune tissue state. Its specific contribution is methodological and cross-dataset: it shows that a compact macrophage/APC programme can be projected and recovered across three independent bulk cohorts rather than inferred from one dataset alone.

The programme also agrees with macrophage-focused RPL literature, but it should be interpreted carefully. Reviews and primary studies have emphasized that decidual macrophages cannot be reduced to a simple M1/M2 dichotomy and may adopt hybrid, context-dependent states [18,22,23]. CCR1+ macrophage dysfunction, M1-like macrophage polarization and impaired anti-inflammatory macrophage programming have been reported in RPL or recurrent spontaneous abortion contexts [19-21]. The present signature includes antigen-presentation genes, inflammatory myeloid markers and C1q-associated macrophage genes, so it is coherent with these reports. However, the bulk projection design cannot determine whether the signal reflects more macrophages, a different macrophage state, increased antigen presentation, complement-associated tissue remodelling or secondary response to tissue injury. This is a key difference from functional macrophage studies: our analysis supports recurrence of a programme, not the mechanism that generates it.

Complement-related genes in the primary signature, especially C1QA, C1QB and C1QC, add biological interest. Complement pathways are active at the implantation site and can contribute to inflammation, trophoblast invasion, vascular remodelling and clearance of apoptotic material [24-26]. Recent decidual bioinformatics work has also linked complement/coagulation modules and macrophage abundance to RPL datasets [27]. The similarity is useful because it places the current C1q-containing macrophage/APC programme within a broader complement-inflammation literature. The difference is equally important: the current study does not claim that complement activation causes RPL or that C1q is independently pathogenic. C1q genes may mark macrophage identity, trophoblast/stromal interaction, tissue-remodelling macrophages or local innate immune activation, and these possibilities require spatial and protein-level validation.

Methodologically, the study contributes a transportability-oriented way to reuse public reproductive transcriptomic data. Single-cell datasets provide cellular resolution but are often small, whereas bulk datasets provide independent cohorts but have mixed-cell composition. Pseudobulk summarization respects biological replication in single-cell analysis and reduces false confidence caused by treating cells as independent samples [37,38]. Mean standardized programme scoring is transparent and interpretable, and the use of directional concordance across three cohorts is appropriate for small public datasets where individual P values are unstable [39,40]. This differs from diagnostic classifier studies, including the original macrophage-marker model associated with GSE214607 [30]. The present analysis deliberately avoids claiming a predictive model and instead asks whether the biological programme itself is transportable.

The weaker behaviour of the secondary signatures is informative. If every GSE214607-derived signature shifted similarly across all external datasets, the result could reflect a nonspecific batch or disease-projection artefact. Instead, the decidua immune-stromal programme was positive but weaker, villi/trophoblast-related programmes were modest or mixed, and the macrophage/APC signature was the most consistent. This supports the interpretation that decidua-relevant inflammatory myeloid biology, rather than generic signature projection, drives the primary result. It also reinforces the importance of tissue context: a villi-derived programme should not necessarily transport strongly into decidua-enriched bulk cohorts.

Translationally, the finding should be treated as a prioritization result. It nominates a decidual inflammatory macrophage/APC axis for future validation in larger, better annotated cohorts. Useful follow-up studies would include euploid versus aneuploid loss stratification, gestational-age matching, standardized tissue sampling, histopathology, fetal karyotype, endocrine and immune metadata, and prospective replication. Spatial transcriptomics, multiplex immunofluorescence, imaging mass cytometry or targeted immunohistochemistry could test whether the programme localizes to specific decidual niches and whether it reflects macrophage abundance, activation state, antigen-presentation activity, complement-associated remodelling or interaction with trophoblast and stromal compartments.

Several limitations remain. First, the external datasets are small, so the evidence rests on directionality across cohorts rather than cohort-level statistical validation. Second, public metadata are incomplete, and clinical labels such as RPL, RM, RSA and unexplained RSA may not represent identical biological entities. Third, bulk programme scores cannot distinguish cell abundance from cell state. Fourth, GSE214607 itself is small and disease-enriched, so derived signatures may capture disease-context programmes rather than universal macrophage identities. Fifth, the reference list and biological framing in v2 are broader than v1.1, but the central empirical result is unchanged: a 30-gene macrophage/APC programme shows positive recurrence across three cohorts.

In conclusion, this study identifies a single-cell-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme that recurs directionally across three independent bulk decidua datasets. The revised evidence framework positions the result within RPL clinical burden, maternal-fetal-interface biology, decidual macrophage literature, complement-associated inflammation and public-data transportability methodology. The finding is biologically coherent and methodologically useful, but it remains exploratory and should not be interpreted as causal, diagnostic or clinically actionable without independent spatial, protein-level and prospective validation.

Conclusion

This study provides exploratory evidence that a single-cell-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme recurs directionally across three independent bulk decidua transcriptomic cohorts. The finding supports the use of single-cell-informed transportability analysis to identify reproducible tissue-state programmes in recurrent pregnancy loss. The macrophage/APC axis is biologically coherent and methodologically transportable, but it should not yet be considered causal, diagnostic or clinically actionable.

Data Availability Statement

Figure-caption support files, reproducibility-audit outputs and release documentation supporting manuscript assembly are summarized in Supplementary Table S4.

All datasets analysed in this study are publicly available from GEO or associated public sequence repositories under accession numbers GSE214607, SRP400574, GSE161969, SRP293708, GSE113790, SRP143517 and

253 GSE178535. Derived signatures, curated metadata summaries, projection scores, transportability summaries,
254 figures, manuscript drafts, QC reports and reproducibility documentation are available at
255 <https://github.com/rmaghembe1/rpl-decidua-transportability>. Large raw downloads and regenerable intermediate
256 objects are not redistributed because the source datasets remain publicly available.

257 **Author contributions**

258 Reuben S. Maghembe: conceptualisation, methodology, software, formal analysis, investigation, data curation,
259 visualisation, writing - original draft, writing - review and editing, project administration.

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263 open-source software and publicly accessible data resources.

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267 integrative transcriptomic reanalysis possible.

268 **Competing interests**

269 The author declares no competing interests.

270 **References**

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363 **Supporting information captions**

364 Supplementary Table S1. Dataset metadata and sample inclusion framework. Curated sample metadata, tissue
365 source, disease group, analysis role and inclusion/exclusion status for GSE214607, GSE161969, GSE113790
366 and GSE178535.

367 Supplementary Table S2. GSE214607-derived projection-ready signature gene sets. Full gene lists for the four
368 projection-ready signatures.

369 Supplementary Table S3. Three-cohort signature projection and transportability statistics. Signature scores,
370 gene-overlap metrics, cohort-specific case-minus-control deltas and cross-cohort concordance summaries.

371 Supplementary Table S4. Figure-caption support and reproducibility audit. Figure-source tables, script/output
372 inventory, release-manifest notes and interpretation-boundary documentation.

373 Supplementary Figure S1. Dataset processing, quality-control and signature-coverage summary.

374 Supplementary Figure S2. Intermediate single-cohort and two-cohort projection summaries supporting the final
375 three-cohort analysis.

376 **Figure legends**

377 Figure 1. Study design and analytical workflow

378 Single-cell-informed transcriptomic transportability workflow for recurrent pregnancy loss decidua analysis.
379 GSE214607 was used as a disease single-cell/pseudobulk reference and GSE161969, GSE113790 and
380 GSE178535 were used as independent bulk decidua projection cohorts. Candidate programmes were derived
381 from pseudobulk contrasts, filtered for interpretability and projected into each bulk cohort using within-cohort
382 standardized expression. Transportability was evaluated by coverage, case-minus-control deltas and directional
383 concordance.

384 Figure 2. GSE214607-derived projection-ready tissue-state signatures

385 Derivation of projection-ready recurrent-miscarriage and tissue-context programmes from GSE214607. Four
386 compact signatures were retained: recurrent-miscarriage decidual inflammatory macrophage/APC up, decidua
387 immune-stromal identity, villi trophoblast/epithelial identity and recurrent-miscarriage villi
388 structural/trophoblast.

389 Figure 3. Three-cohort transportability of the macrophage/APC programme

390 The recurrent-miscarriage decidual inflammatory macrophage/APC programme showed complete 30/30 gene
391 coverage and positive case-associated shifts in GSE161969, GSE113790 and GSE178535. Deltas were +1.019,
392 +0.282 and +0.941, respectively.

393 Figure 4. Cross-cohort delta summary for all projected signatures

394 Case-minus-control programme-score deltas for four GSE214607-derived signatures across external cohorts.
395 The macrophage/APC programme showed the strongest and most consistent positive directional recurrence;
396 villi/trophoblast-associated programmes were weaker or mixed.

397 **Tables**

398 Table 1. Public datasets used in the analysis

Dataset	Assay/data type	Tissue/source	Groups used	Role in study	Key processing	Main limitation
GSE214607	Single-cell RNA-seq / pseudobulk reference	First-trimester decidua and villi	Control and recurrent miscarriage	Disease single-cell reference; programme derivation	Raw 10x matrices imported, QC-filtered and aggregated to pseudobulk profiles	Limited sample numbers; exploratory programme derivation
GSE161969	Bulk RNA-seq processed expression	Decidua	Control and RPL	External projection cohort	Gene-symbol FPKM matrix; log2(FPKM + 1); within-cohort scoring	Very small cohort; limited metadata
GSE113790	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	FPKM matrix imported; labels curated; within-cohort scoring	Very small cohort; metadata harmonization constraints
GSE178535	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	Public normalized log2 matrix; labels curated; within-cohort scoring	Very small cohort; platform/preprocessing differences

399

400 Table 2. GSE214607-derived projection-ready signatures

Code	Short label	Source contrast	Genes	Biological interpretation	Use
M/APC-RM	RM decidua inflammatory macrophage/APC up	RM decidua vs control decidua	30	Antigen-presentation, inflammatory myeloid and complement-associated macrophage/APC programme	Primary programme
DEC-IS	Decidua immune-stromal identity	Decidua vs villi	30	Decidua-enriched immune and stromal tissue-context programme	Secondary comparator
VIL-TE	Villi trophoblast/epithelial identity	Villi vs decidua	30	Villi/trophoblast/epithelial tissue-context programme	Tissue-context comparator
VIL-ST-RM	RM villi structural/trophoblast programme	RM villi vs control villi	30	Villi structural/trophoblast-associated programme	Secondary tissue-context comparator

401

402 Table 3. Three-cohort transportability summary

Programme	GSE161969 delta	GSE113790 delta	GSE178535 delta	Positive cohorts	Minimum coverage	Concordance interpretation
RM decidua macrophage/APC	+1.019	+0.282	+0.941	3/3	100%	Concordant positive directional signal
Decidua immune- stromal	+0.573	+0.206	+0.031	3/3	96.7%	Positive but weaker/context- dependent
Villi trophoblast/ epithelial	+0.134	+0.406	+0.156	3/3	100%	Modest positive; tissue- context dependent
RM villi structural/ trophoblast	-0.097	+0.245	-0.007	1/3	96.7%	Mixed/weak; not transportable in decidua cohorts

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